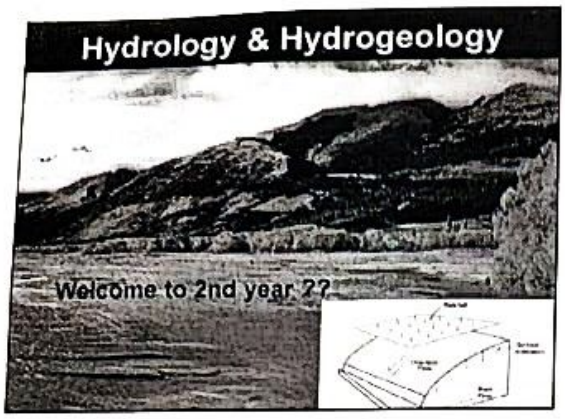
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Course details

- Introduction; hydrologic cycle, precipitation, run off, evapotranspiration and their measurements, hydrograph construction and interpretation, drainage basin evaluation and its quantitative evaluation, drainage patterns, velocity of river flow, stage and discharge measurements; flood.
- Introduction, origin and occurrence of ground water, rock properties affecting ground water, subsurface distribution of groundwater, geological formations as aquifers, types of aquifers. Ground water movement and groundwater modeling. Fundamentals of well hydraulics, steady and unsteady flow, equilibrium and non-equilibrium equations, well losses, well interference. Ground water exploration, determination of aquifer characteristics and pumping test. Detailed study of the different types of aquifer in Bangladesh and its problems. Water wells, well design and well development.

Textbook and/or other Required Material

- Engineering Hydrology- K. Subramanya, 3rd edition
- Hydrology-principles, analysis and design- H.M Raghunath
- Hydrology-water quality and quality control-M. Wanielista et.al.
- Groundwater Hydrology- D.K. Todd-3rd edition.
- Schwartz, F.W., and H. Zhang, Fundamentals of Ground Water. Wiley, 2003.
- *Bedient & Huber, "Hydrology and Floodplain Analysis", Third Edition, Prentice Hall, 2002.*
- Excerpt from Ven Te Chow, D.R. Maidment, L.W. Mays, Applied Hydrology, McGraw Hill.
- Excerpt from W. Viessman, Jr., G. L. Lewis, J. W. Knapp, Introduction to Hydrology, Harper Collins Publishers.

Course Objectives

1. Provide a broad understanding of a wide range of topics in hydrology of importance to people and their environment.
2. Learn basic principles that govern the hydrological processes.
3. Apply the knowledge to solve practical hydrologic problems.

Course Outcomes

1. Understand hydrologic cycle.
2. Understand the hydrologic processes of precipitation, evaporation, infiltration, snow and snowmelt, and stream flow. Have the ability to conduct all the relevant calculations related to these processes.
3. Have the ability to conduct rainfall runoff analysis using the unit hydrograph approach.

Course Outcomes

4. Understand the importance of frequency analysis in engineering design. Have the ability to conduct various flood frequency analysis.
5. Have the ability to conduct flood routing through reservoirs and rivers.
6. Have the ability to conduct hydrologic model simulations using HEC-HMS.
7. Understand the key characteristics and approaches in urban hydrology.

Overview

- **Introduction to Hydrology**
- Hydrologic Cycle and its Elements
- **Water Budget Equation**
- **Precipitation**
- **Evaporation and ET**
- **Infiltration Loss**
- **Stream flow and the Hydrograph**
- **Hydrologic Measurement**

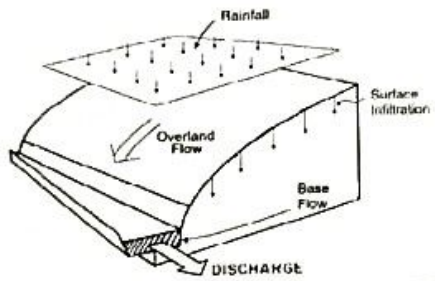
The Fate of Water

- **Evaporation:** Transformation of liquid water to vapor.
- **Transpiration:** Vegetation releasing water to the atmosphere.
- **Condensation:** transformation of vapor to liquid water.
- **Precipitation:** Liquid or solid water falling on the earth's surface.
- **Surface runoff:** Movement of water on the Earth's Surface.
- **Infiltration:** water penetrating into the ground to become groundwater.
- **Interception:** Precipitation kept from reaching the ground.
- **Groundwater flow:** flow of water in the sub-surface.

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Hydrology

- Rainfall – Runoff Process



1.1 Introduction to Hydrology

Hydrology

from Greek: Ὕδωρ, hudôr, "water";
and Λόγος, logos, "knowledge"

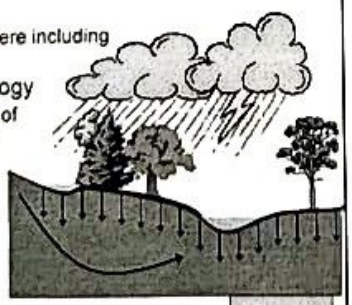
Hydrology is a multidisiplinary natural science that deals with the occurrence, properties, distribution and movement of surface and ground – water on the earth (in the natural and man-made environments). In a general sense, it is the science which deals with the various phases of the hydrologic cycle.

Hydrology is basically an applied science. To further emphasis the degree of applicability, the subject is sometimes classified as-

1. Scientific Hydrology- the study which is concerned chiefly with academic aspects
2. Engineering or applied hydrology- a study concerned with engineering applications.

Hydrology

- Meteorology
  - Study of the atmosphere including weather and climate
- Surface water hydrology
  - Flow and occurrence of water on the surface of the earth
- Hydrogeology
  - Flow and occurrence of ground water



Watersheds 1

Main Branches

HYDROLOGY

Surface Water Hydrology

Ground Water Hydrology

Introduction to Hydrology

Why is hydrology important?

- **Water is essential to life**
- **Humans have been concerned with managing water as a necessity of life and as a potential hazard at least since the first civilizations developed along the banks of rivers**
- **Water is the most precious resource**
- **Knowledge of the storage and movement of water is necessary for understanding all environmental processes (and problems)**

Introduction to Hydrology

- Applications of Hydrology
  - Designing dams for water supply or hydroelectric power generation;
  - Designing sewers and urban drainage system;
  - Designing bridges;
  - Mitigating and predicting flood, landslide and drought risk;
  - Providing drinking water;
  - Designing irrigation schemes and managing agricultural productivity;
  - Determining the water balance of a region.

Introduction to Hydrology

Knowledge of hydrology is necessary for:

- **Water resources management**
- **Understanding the impacts of land use and climate change on water quantity and quality**
- **Understanding stream quality issues (related fisheries/ecology)**
- **Flood prediction and dam management**
- **Understanding aquifer pollution and remediation solutions**

SCOPE OF HYDROLOGY

The study of hydrology helps us to know

(i) the maximum probable flood that may occur at a given site and its frequency; this is required for the safe design of drains and culverts, dams and reservoirs, channels and other flood control structures.

(ii) the water yield from a basin—its occurrence, quantity and frequency, etc; this is necessary for the design of dams, municipal water supply, water power, river navigation, etc.

(iii) the ground water development for which a knowledge of the hydrogeology of the area, i.e., of the formation soil, recharge facilities like streams and reservoirs, rain-fall pattern, climate, cropping pattern, etc. are required.

(iv) the maximum intensity of storm and its frequency for the design of a drainage project in the area.

HYDROLOGICAL DATA

For the analysis and design of any hydrologic project adequate data and length of records are necessary. A hydrologist is often posed with lack of adequate data.

The basic hydrological data required are:

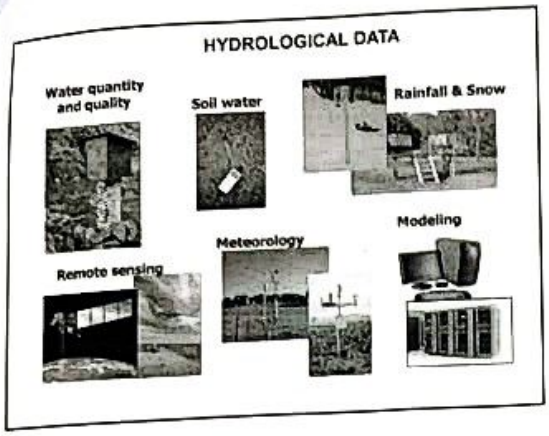
(i) Climatological data
(ii)Hydrometeorological data like temperature, wind velocity, humidity, etc.
(iii) Precipitation records
(iv) Stream-flow records
(v)Seasonal fluctuation of ground water table or piezometric heads
(vi) Evaporation data
(vii) Cropping pattern, crops and their consumptive use
(viii) Water quality data of surface streams and ground water

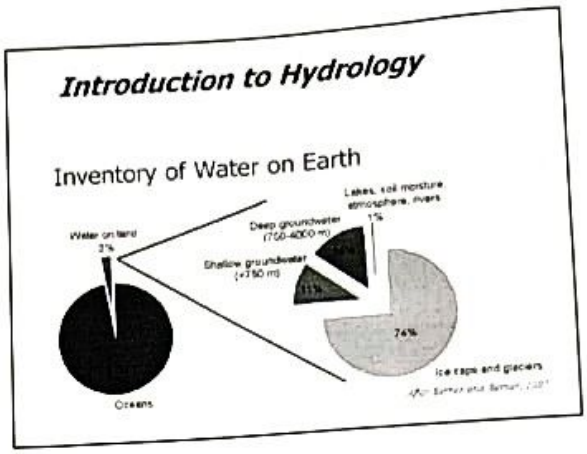
HYDROLOGICAL DATA

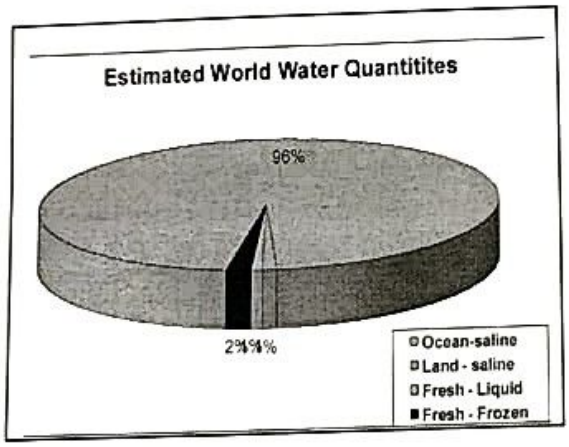
(ix) Geomorphologic studies of the basin, like area, shape and slope of the basin, mean and median elevation, mean temperature (as well as highest and lowest tempera-ture recorded) and other physiographic characteristics of the basin; stream density and drainage density; tanks and reservoirs.

(X) Hydrometeorological characteristics of basin:

(i) long term precipitation, space average over the basin using isohyets and several other methods (Rainbird, 1968)
(ii) Depth-area-duration (DAD) curves for critical storms (station equipped with self-recording raingauges).
(iii) Isohyetal maps—Isohyets may be drawn for long-term average, annual and monthly precipitation for individual years and months
(iv) Cropping pattern—crops and their seasons
(v) Daily, monthly and annual evaporation from water surfaces in the basin
(vi) Water balance studies of the basin
(vi) Soil conservation and methods of flood control

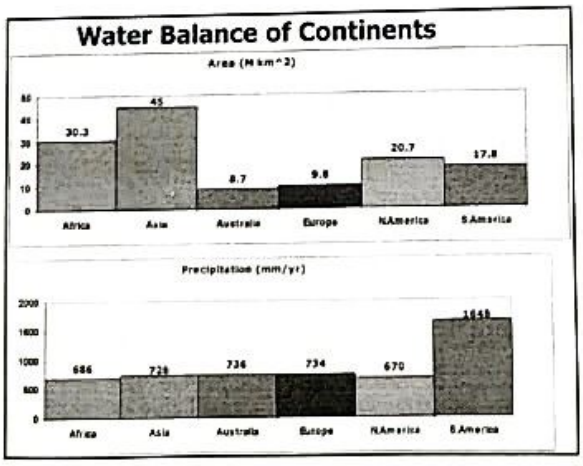


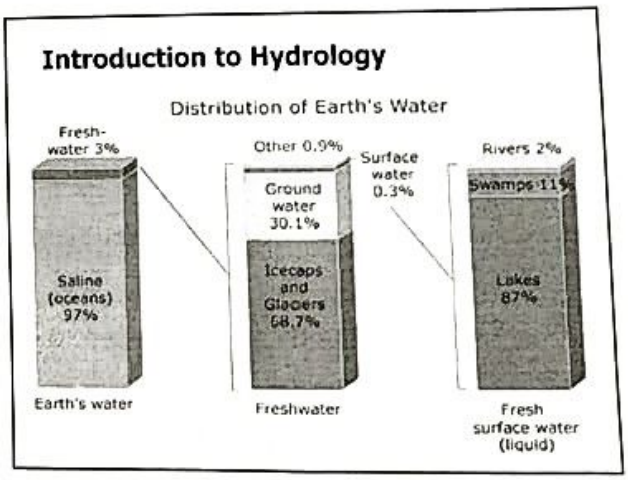


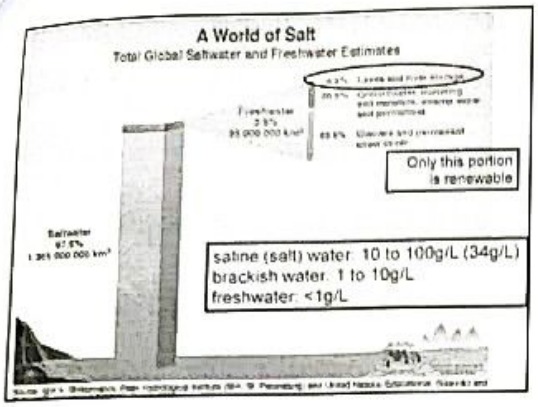


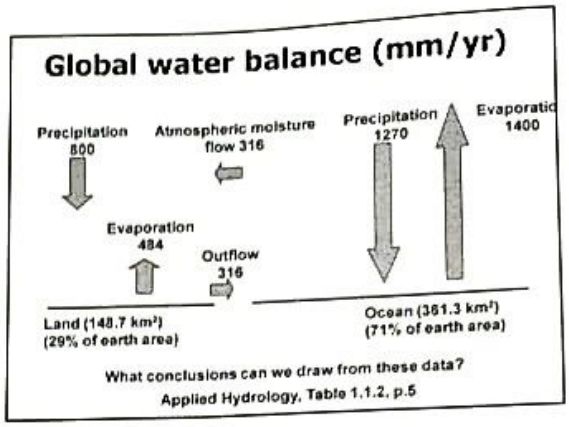
Global annual water balance

SN	Item	Ocean	Land
1	Area (km²)	361.3	148.8
2	Precipitation (km³/year)	458,000	119,000
	(mm/year)	1270	800
3	Evaporation (km³/year)	505,000	72,000
	(mm/year)	1400	484
4	Runoff to ocean		
	Rivers (km³/year)		44,700
	Groundwater (km³/year)		2,200
	Total Runoff (km³/year)		47,000
	(mm/year)		316









1.1 Introduction to Hydrology

A few numbers everyone should know:

- 1.1 billion people lack access to safe water, roughly 1/6 of the world's population.
- One flush of a Western toilet uses as much water as the average person in the developing world uses for a whole day's washing, drinking, cleaning and cooking.
- Some 6000 children die every day from diseases associated with unsafe water and poor sanitation = 20 jumbo jets crashing every day.
- Floods affected more than 75% of all people impacted by natural disasters during the 1990s and caused over 33% of the total estimated costs of natural disasters.

1.2 Hydrologic Cycle

- **Water exists on the earth in all its three states, viz. liquid, solid, gaseous and in various degrees of motion.**



1.2 Hydrologic Cycle

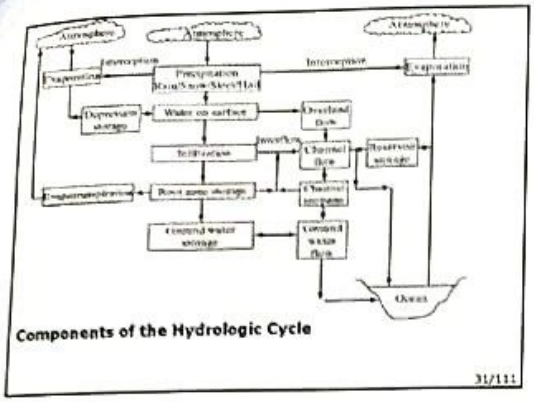
In a global sense, the occurrence, distribution and movement of water in the natural environment can be visualized through a cyclic process known as the hydrologic cycle.

- The hydrologic cycle is the central focus of hydrology.
- The cycle has no beginning or end.
- Its processes occur continuously.
- The hydrological cycle can be thought of in terms of Storages (the atmosphere, land, and the oceans), and the transfer mechanisms that move water from one storage to another.
- Transfer of water from one storage to another is driven by energy, either thermal (from the sun) or mechanical (due to gravity).

Components of hydrologic cycle

- **Precipitation**
  - rain, snow, fog interception
- **Runoff**
  - surface, subsurface
- **Storage**
- **Evaporation**
  - soil, plants, water surface

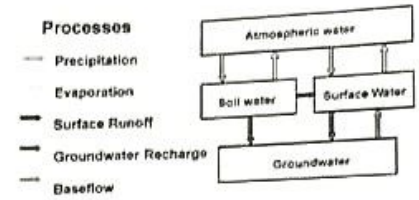
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Components of the Hydrologic Cycle

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Hydrologic Cycle Processes



Hydrologic Parameters

- Precipitation intensity & duration for design
- Infiltration rate (watershed soil type and moisture condition)
- Watershed surface cover – overland roughness
- Watershed drainage network geometry
- Watershed slope
- Time of concentration

1.2 Hydrologic Cycle

Hydrologic System

It is common in engineering analysis to represent the different components of the hydrologic cycle as components of a complex system.

Definition of a System

System: a set of interconnected parts or components that form a whole:

- Have some structure or organization
- Have functional relationships

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1.2 Hydrologic Cycle

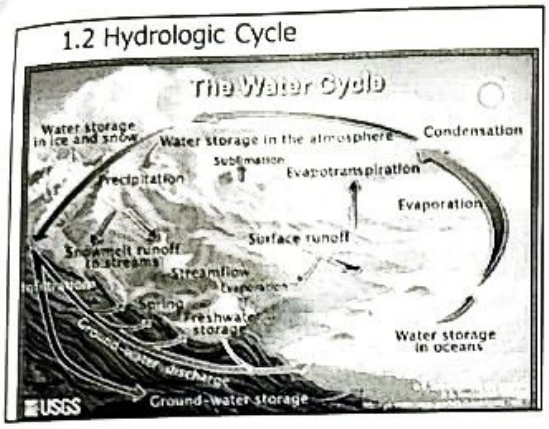
- Hydrologic cycle is the water transfer cycle, which occurs continuously in nature; the three important phases of the hydrologic cycle are: (a) Evaporation and evapotranspiration (b) precipitation and (c) runoff and is shown in Fig.
- The globe has one-third land and two-thirds ocean. Evaporation from the surfaces of ponds, lakes, reservoirs, ocean surfaces, etc. and transpiration from surface vegetation i.e., from plant leaves of cropped land and forests, etc. take place. These vapors rise to the sky and are condensed at higher altitudes by conden-sation nuclei and form clouds, resulting in droplet growth. The clouds melt and sometimes burst resulting in precipitation of different forms like rain, snow, hail, sleet, mist, dew and frost. A part of this precipitation flows over the land called runoff and part infilters into the soil which builds up the ground water table.

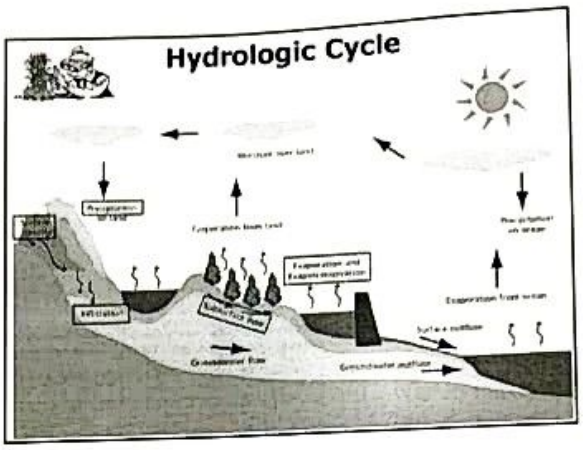
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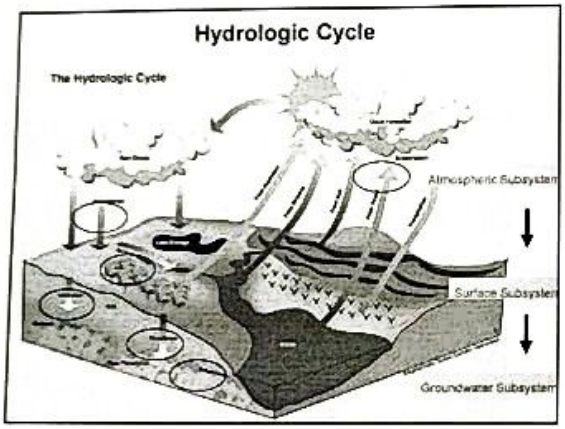
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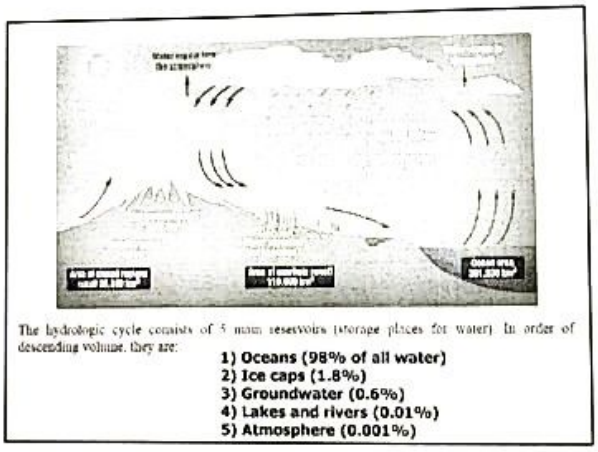
1.2 Hydrologic Cycle

- The surface runoff joins the streams and the water is stored in reservoirs. A portion of surface runoff and ground water flows back to ocean. Again evaporation starts from the surfaces of lakes, reservoirs and ocean, and the cycle re- peats. Of these three phases of the hydrologic cycle, namely, evaporation, precipitation and runoff, it is the 'runoff phase', which is important to a civil engineer since he is concerned with the storage of surface runoff in tanks and reservoirs for the purposes of irrigation, municipal water supply hydroelectric power etc



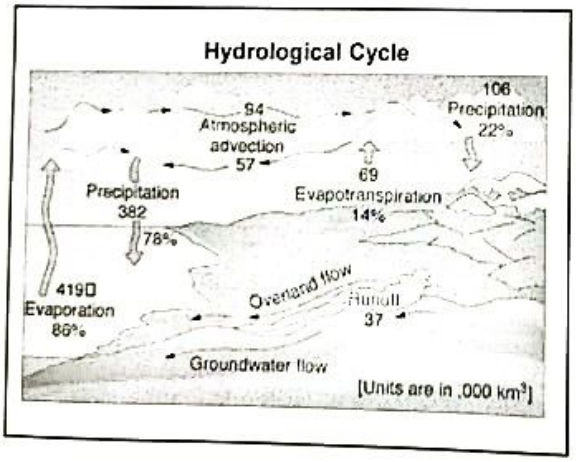


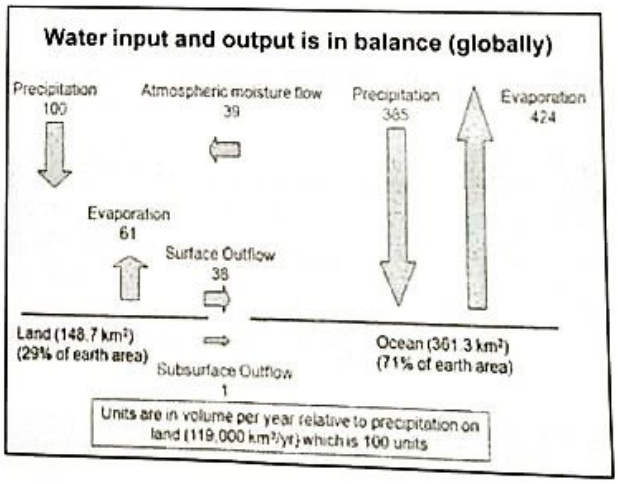




The hydrologic cycle consists of 5 main reservoirs (storage places for water). In order of descending volume, they are:

1) **Oceans (98% of all water)**
2) **Ice caps (1.8%)**
3) **Groundwater (0.6%)**
4) **Lakes and rivers (0.01%)**
5) **Atmosphere (0.001%)**





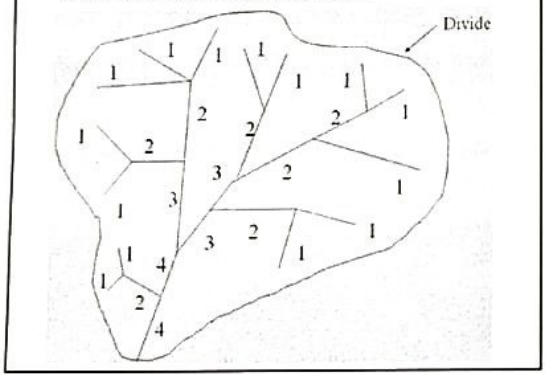
Catchment and Basin

- A *Catchment* is a portion of the earth's surface that collects runoff and concentrates it at its furthest downstream point, referred to as the catchment outlet.
- The runoff concentrated by a catchment flows either into a larger catchment or into the ocean.
- The place where a stream enters a larger stream or body of water is referred to as the *mouth*.
- The terms watershed and basin are commonly used to refer to catchments. Generally, *watershed* is used to describe a small catchment (stream watershed), whereas *basin* is reserved for large catchments (river basins).

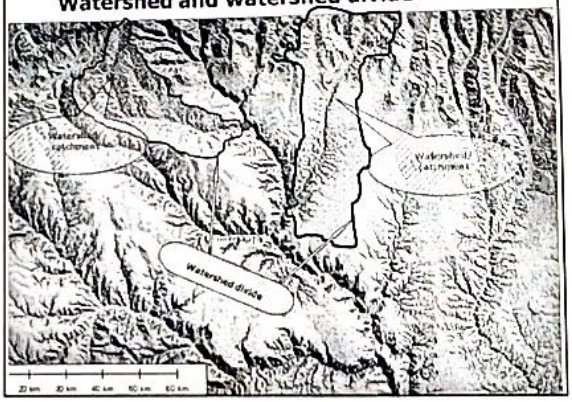
Watershed and Stream order

- The watershed or basin is defined by the surrounding topography, the perimeter of which is called a *divide*. It is the highest elevation surrounding the watershed. All of the water that falls on the inside of the divided has the potential to be shed into the streams of the basin encompassed by the divide. Water falling outside of the divide is shed to another basin.
- The water flowing in streams is called *stream flow*
- Horton suggested a classification of stream order as a measure of the amount of branching within a basin. A first order stream is a small, un-branched tributary. A second order stream has only first order tributaries. A third order stream has only first and second order tributaries and so on. When a channel of lower order joins a channel of higher order, the channel downstream retains the higher of the two orders.

Watershed and Stream order



Watershed and watershed divide



1.3 Water Budget Equation

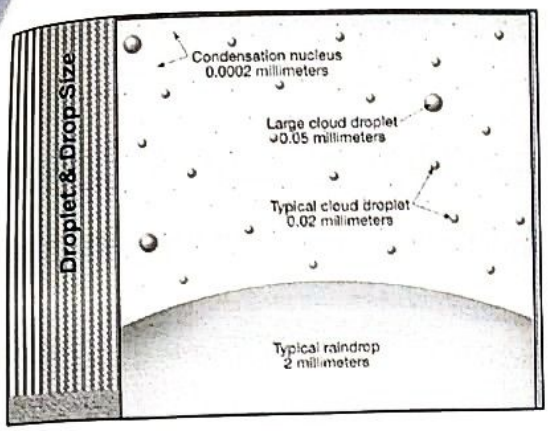
Catchment area

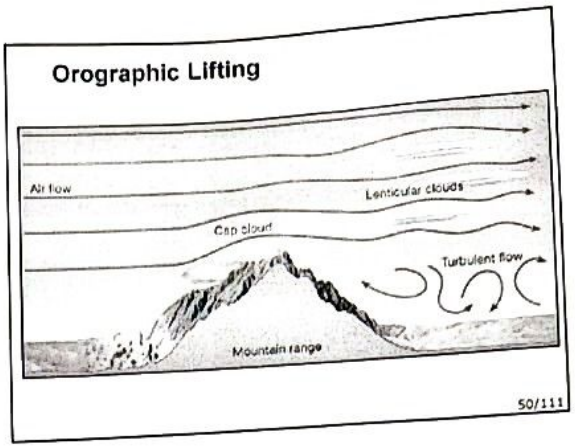
- The area of land draining in to a stream or a water course at a given location is called catchment area / drainage area / drainage basin / watershed.
- A catchment area is separated from its neighbouring areas by a ridge called divide / watershed.

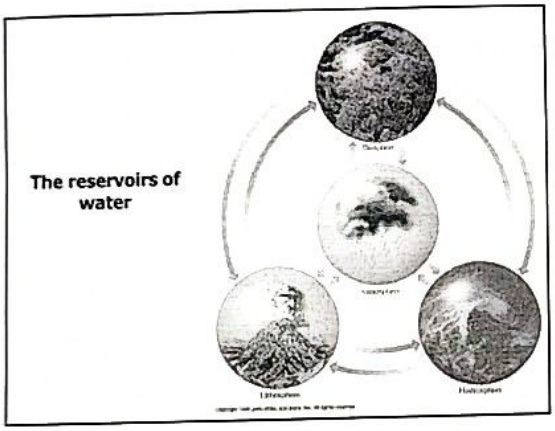
1.3 Water Budget Equation

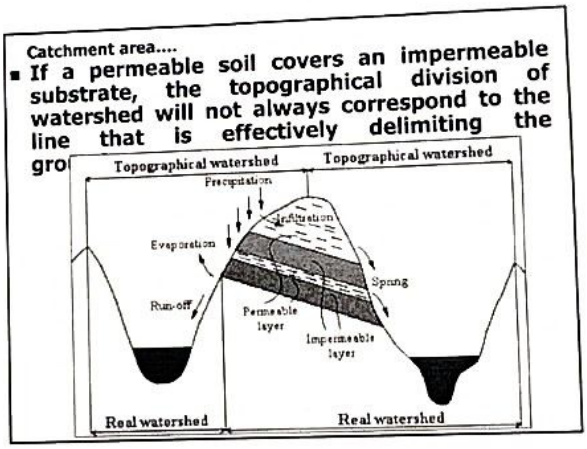
Catchment area....

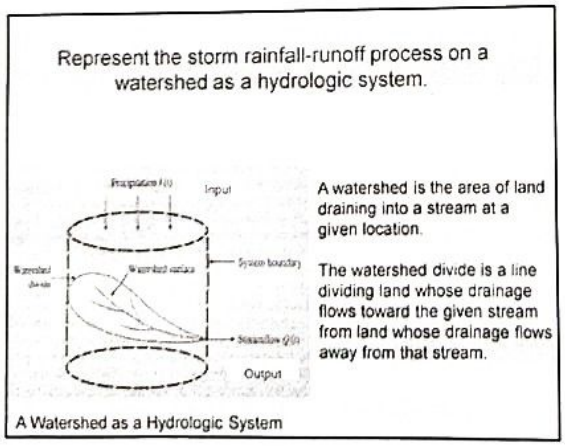
- A watershed is a geographical unit in which the hydrological cycle and its components can be analysed. The equation is applied in the form of water-balance equation to a geographical region, in order to establish the basic hydrologic characteristics of the region. Usually a watershed is defined as the area that appears, on the basis of topography, to contribute all the water that passes through a given cross section of a stream.











HYDROLOGIC EQUATION

- The hydrologic equation is simply the statement of the law of conservation of matter and is given by
- $I = O + \Delta S$
  - where I = inflow, O = outflow, ΔS = change in storage
- This equation states that during a given period, the total inflow into a given area must equal the total outflow from the area plus the change is storage. While solving this equation, the ground water is considered as an integral part of the surface water and it is the subsurface inflow and outflow that pose problems in the water balance studies of a basin.

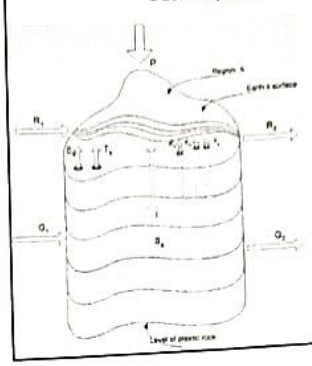
Basic Equation of Hydrologic Cycle

Input (Rainfall) → Process (Basin) → Output (Runoff)

Simple Hydrologic System Model

Unsteady Flow Equation ; I-Q=dS/dt
I = Input (volume/time)
O= Output (volume/time)
dS/dt = Time rate of change of storage

Basic Equation of Hydrologic Cycle



P=Precipitation
E=Evaporation
T=Transpiration
R=Surface Runoff
G=Groundwater Flow
Rg=Subsurface Flow
I=Infiltration
S=Storage

s=Land Surface
g=Groundwater

Basic Equation of Hydrologic Cycle

Water Budget in Land Surface
(P+R1+Rg)-(R2+Es+Ts+I)=ΔSs (1)

Water Budget in Groundwater
(I+G1)-(G2+Rg+Eg+Tg)=ΔSg (2)

P-(R2-R1)-(Es+Eg)-(Ts+Tg)-(G2-G1)= ΔSs+ΔSg *

R (Net Surface Flow)=R2-R1
E (Net Evaporation) =E2+E1
T (Net Transpiration)=Ts+Tg
G (Net Groundwater Flow)=G2-G1
ΔS= ΔSs+ΔSg

P-R-E-T-G=ΔS

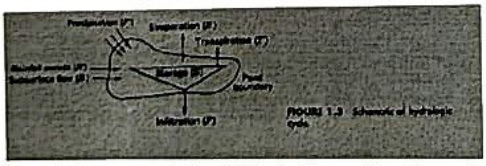
The Water Budget

The water budget represents the inventory of water for a specific water body (or hydrologic region) during a certain time interval.

$$P - R - G - E - T = \Delta S$$

- P: precipitation, [unit of height] or [unit of volume / unit of time]
- R: runoff, [unit of height] or [unit of volume / unit of time]
- G: groundwater flow, [unit of height] or [unit of volume / unit of time]
- E: evaporation, [unit of height] or [unit of volume / unit of time]
- T: transpiration, [unit of height] or [unit of volume / unit of time]
- ΔS: change in storage, [unit of height] or [unit of volume / unit of time]

Uses of the hydrologic cycle (HC)



- One of the uses of the HC is in the estimation of surface storage.
- Storing and transferring a sufficient quantity of water has been one of the major problems.
  - What volume of water is stored in a surface reservoir/soil and how does the volume change over time? What causes the water supply to be depleted or increased?
  - How are the storage and releases managed?

What to do about units?

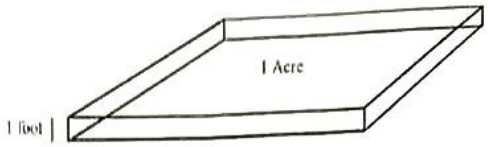
- Rainfall is expressed in mm, in
- Stream flow is expressed in cubic feet/cubic meter per second/minute
- Evapotranspiration is expressed in mm.
- Conversion

Water Depth

- We have to use the same units; thus we have to remove the area from our calculation
- We need to convert volume into unit depth; thus what's water depth:

Water depth (d) = Volume of water (V) / Surface of the field (A)

Conversion



1 acre foot

43,560 cubic feet

348,480 gallons

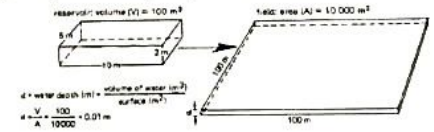
1 acre-foot = 1317.25 m^3

Problem 1

- Suppose there is a reservoir, filled with water, with a length of 5 m, a width of 10 m and a depth of 2 m. All the water from the reservoir is spread over a field of 1 hectare. Calculate the water depth (which is the thickness of the water layer) on the field.

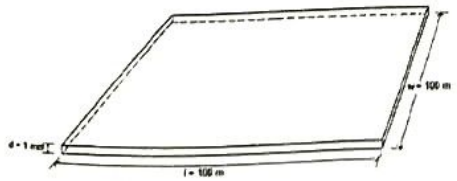
Answer 1

- Surface of the field = 10 000 m2
  Volume of water = 100 m3
- Formula:
  d = v/a =100 / 10,000 = 0.01 m = 10 mm



Problem 2

- A water layer 1 mm thick is spread over a field of 1 ha. Calculate the volume of the water (in m^3),



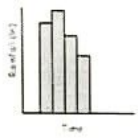
Answer 2

- <u>Given</u>
- Surface of the field = 10 000 m2
  Water depth = 1 mm =1/1 000 = 0.001 m
- Formula: Volume (m^3) = surface of the field (m^2) x water depth (m)
- <u>Answer</u>
  V = 10 000 m^2 x 0.001 m
  V = 10 m^3 or 10 000 liters

1.2 Hydrologic Cycle

Rainfall Hyetograph

The total input into a watershed associated with a particular storm event can be represented graphically through a plot of rainfall versus time.



Runoff Hydrograph

The total output from a watershed in response to a particular input (precipitation) can be represented graphically through a plot of flow vs time.



